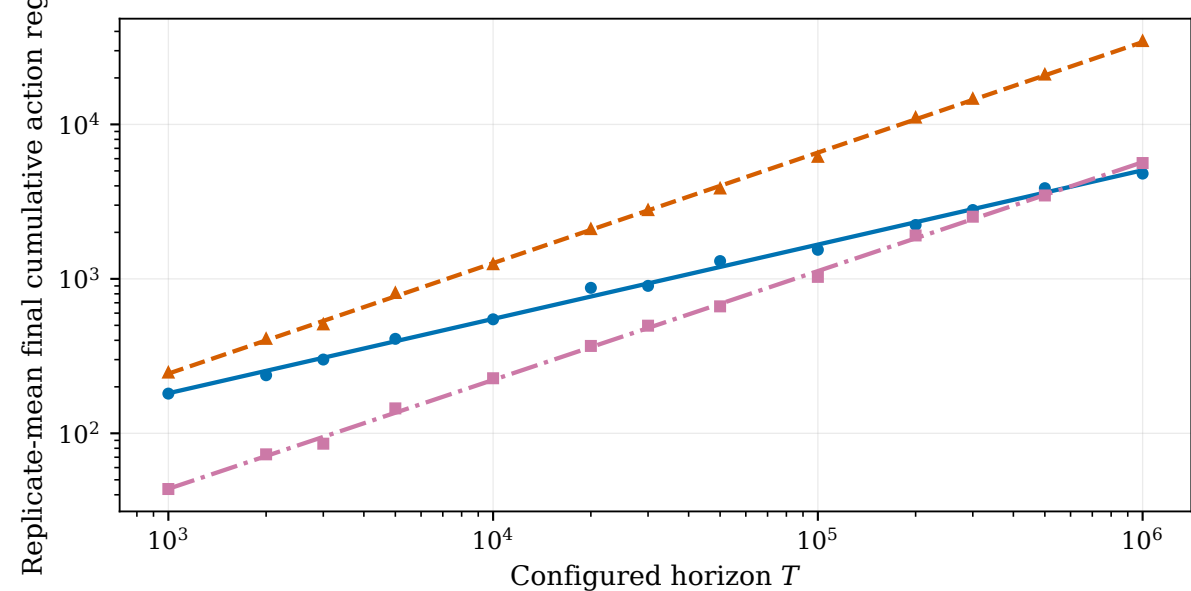


Experiment Information

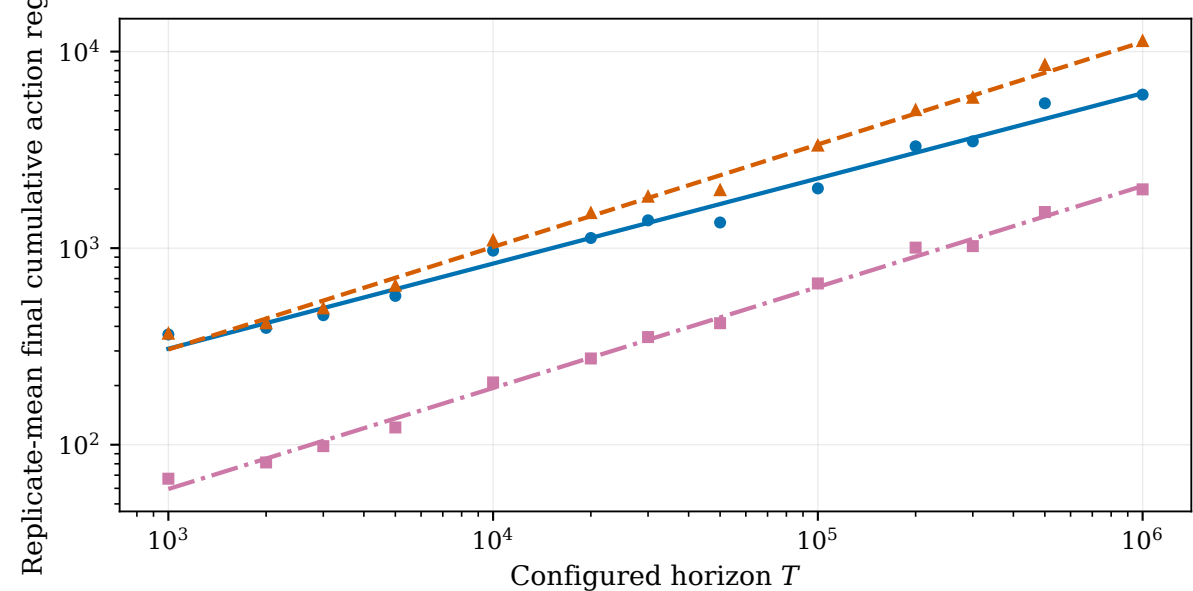
Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithm: EXP3
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.004803, SD = 0.00170114, SE = 0.000491078
Internal R/T: mean = 0.00562192, SD = 0.00148798, SE = 0.000429542
Swap R/T: mean = 0.0346296, SD = 0.00799226, SE = 0.00230717
External: $\alpha = 0.481$, $c = 6.544$, $R^2 = 0.997$
Internal: $\alpha = 0.705$, $c = 0.3356$, $R^2 = 0.999$
Swap: $\alpha = 0.716$, $c = 1.737$, $R^2 = 1.000$



- External: $\alpha = 0.481, c = 6.544$
- Internal: $\alpha = 0.705, c = 0.3356$
- Swap: $\alpha = 0.716, c = 1.737$

Experiment Information

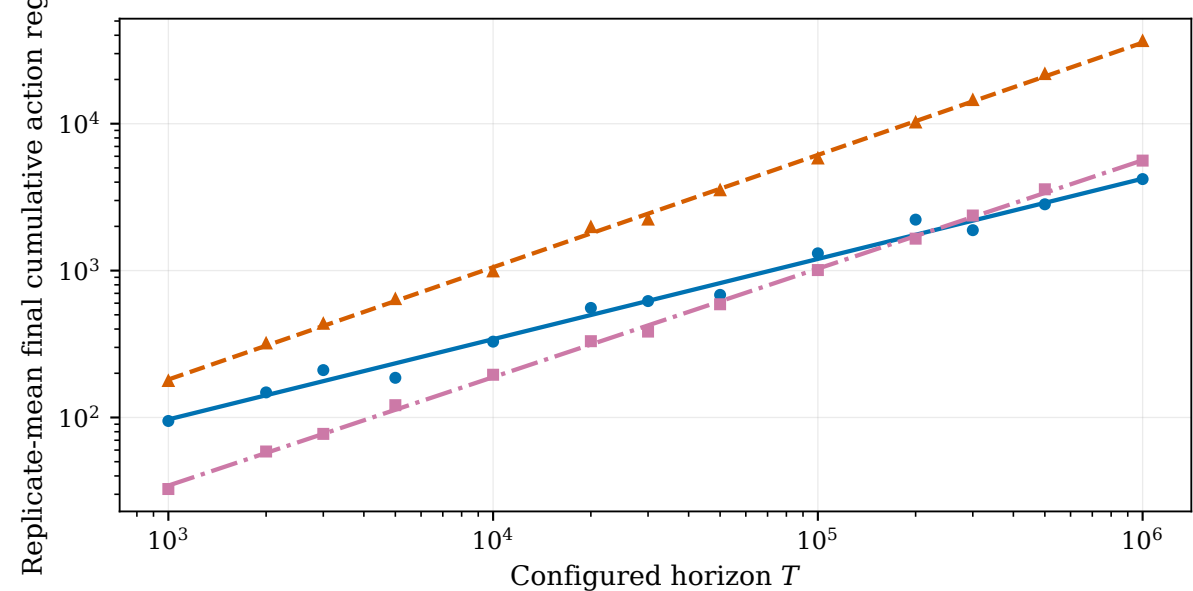
Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.00603992, SD = 0.00179243, SE = 0.000517429
Internal R/T: mean = 0.00199142, SD = 0.000436228, SE = 0.000125928
Swap R/T: mean = 0.0113095, SD = 0.00147638, SE = 0.000426195
External: $\alpha = 0.433$, $c = 15.43$, $R^2 = 0.985$
Internal: $\alpha = 0.514$, $c = 1.71$, $R^2 = 0.996$
Swap: $\alpha = 0.522$, $c = 8.314$, $R^2 = 0.994$



- External: $\alpha = 0.433, c = 15.43$
- Internal: $\alpha = 0.514, c = 1.71$
- Swap: $\alpha = 0.522, c = 8.314$

Experiment Information

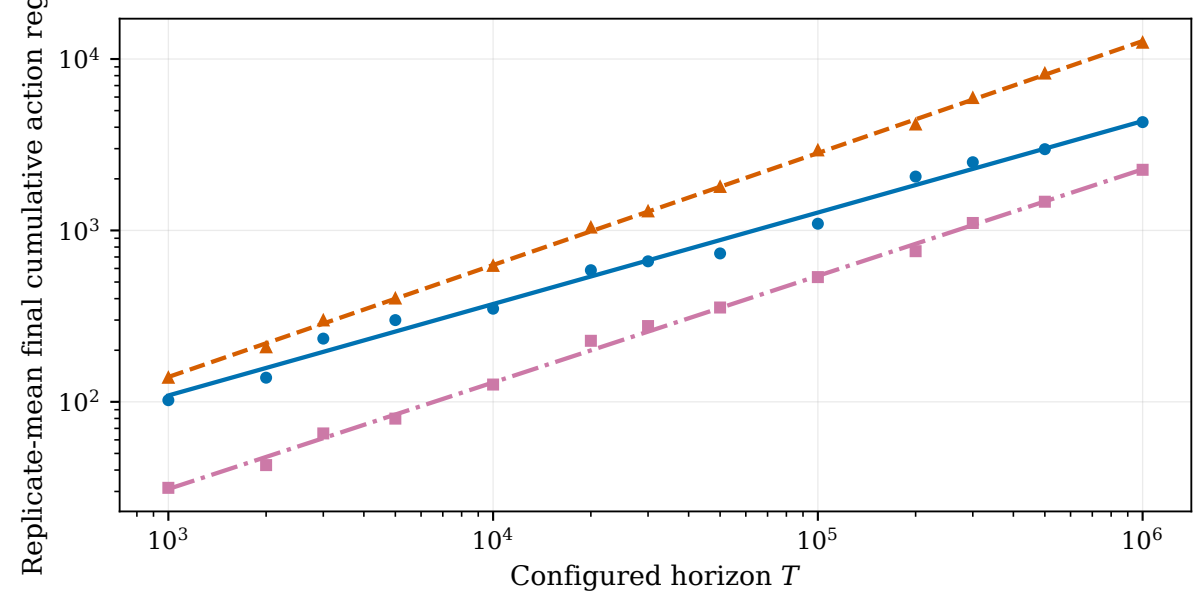
Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithm: EXP3-IX
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.00419725, SD = 0.0025064, SE = 0.000723535
Internal R/T: mean = 0.00560775, SD = 0.00108382, SE = 0.000312871
Swap R/T: mean = 0.036513, SD = 0.00761423, SE = 0.00219804
External: $\alpha = 0.546$, $c = 2.228$, $R^2 = 0.988$
Internal: $\alpha = 0.738$, $c = 0.2098$, $R^2 = 0.999$
Swap: $\alpha = 0.764$, $c = 0.9228$, $R^2 = 0.999$



- External: $\alpha = 0.546$, $c = 2.228$
- · - Internal: $\alpha = 0.738$, $c = 0.2098$
- - Swap: $\alpha = 0.764$, $c = 0.9228$

Experiment Information

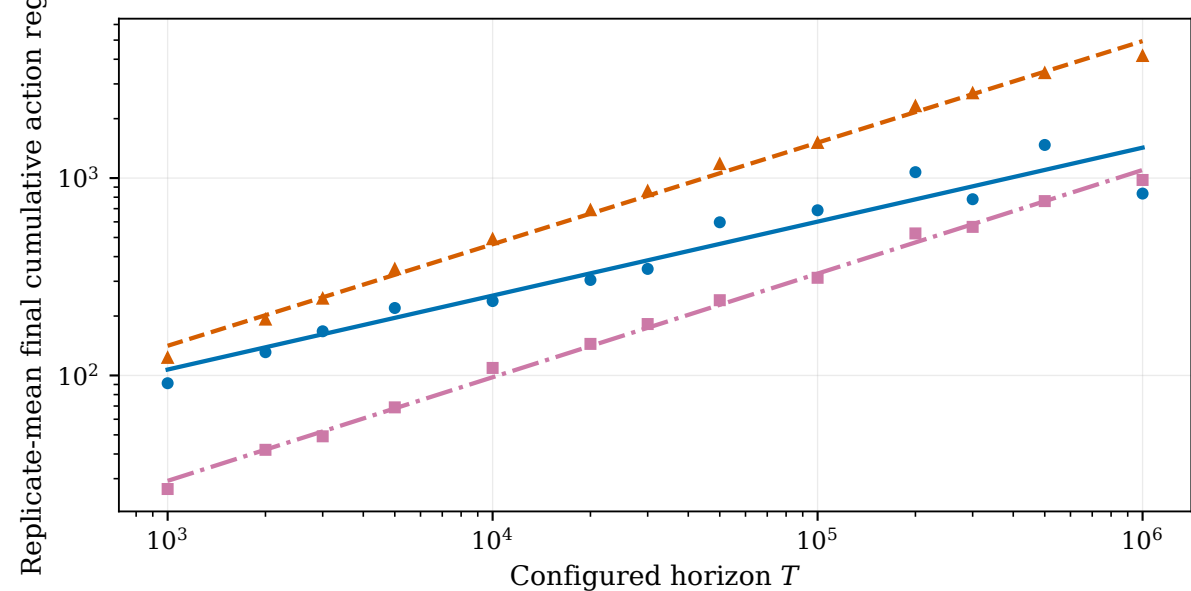
Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.0042845, SD = 0.00121424, SE = 0.000350521
Internal R/T: mean = 0.00225817, SD = 0.000532163, SE = 0.000153622
Swap R/T: mean = 0.0124485, SD = 0.00251438, SE = 0.000725839
External: $\alpha = 0.534$, $c = 2.731$, $R^2 = 0.990$
Internal: $\alpha = 0.622$, $c = 0.423$, $R^2 = 0.998$
Swap: $\alpha = 0.653$, $c = 1.532$, $R^2 = 0.999$



- External: $\alpha = 0.534, c = 2.731$
- . - Internal: $\alpha = 0.622, c = 0.423$
- - Swap: $\alpha = 0.653, c = 1.532$

Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Historical-frequency
Actions: 10
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.000834583, SD = 0.00119436, SE = 0.000344781
Internal R/T: mean = 0.000978167, SD = 0.000230327, SE = 6.64896e-05
Swap R/T: mean = 0.00415367, SD = 0.00111991, SE = 0.000323289
External: $\alpha = 0.375$, $c = 8.049$, $R^2 = 0.929$
Internal: $\alpha = 0.525$, $c = 0.7781$, $R^2 = 0.997$
Swap: $\alpha = 0.515$, $c = 4.04$, $R^2 = 0.995$



- External: $\alpha = 0.375$, $c = 8.049$
- Internal: $\alpha = 0.525$, $c = 0.7781$
- Swap: $\alpha = 0.515$, $c = 4.04$